

Elevator Maintenance Platform

AI-Assisted Fleet Management & Technical Support System

Developed solely by Erim AKCAN, for his father's elevator installation, repair, and maintenance business.

Portfolio: akcanerim.github.io/portfolio

The Elevator Maintenance Platform is a management system built to replace the notebook-and-phone-call workflow my dad's elevator business was running on. He was getting calls at all hours for breakdowns and people stuck in elevators, with no central record of which buildings were overdue for service or what had already been done. The platform tracks every elevator he services, logs maintenance history, manages pricing, and includes an AI assistant that helps technicians reason through fault codes on site.

Technologies

Python (FastAPI) - backend, REST-style API

SQLAlchemy + SQLite - data layer

OpenAI API (GPT-4o-mini) - AI technical assistant

Vanilla HTML, CSS, JavaScript - frontend (no framework)

Core Features

Elevator & Building Tracking

Every elevator is logged with its serial number, building, address, and maintenance status. The admin dashboard surfaces this as a risk-sorted list, so overdue or high-risk elevators are visible at a glance instead of buried in a spreadsheet.

Maintenance History

Each service visit is recorded against a specific elevator: which tasks were performed (general maintenance, oil change, door adjustment, safety test, and more), who did it, and when. Past records stay attached to the elevator, so the full service history is one click away.

Role-Based Access

Three roles: admin, moderator, and user see different levels of the system. The admin account can manage users, pricing, and site content directly from the interface, without needing to touch the database.

AI Technical Assistant

This is the part I'm most proud of. It's built on GPT-4o-mini with a system prompt that encodes real elevator fault codes (door lock circuit faults, overload warnings, safety circuit errors, photocell faults) and scenario-specific safety instructions for example, telling the user to never force a door stuck between floors, or to call emergency services if someone is unreachable inside the cabin. A technician can type in an error code on site and get an explanation of what it means and what to do next, while it still handles general questions like a normal assistant would.

Pricing & Content Management

Service prices, the vision/mission statement, and the "about us" content are all editable from the admin panel, stored as key-value site content in the database rather than hardcoded, so admin can update prices himself without asking me to change code.

Challenges & Solutions

- **Challenge:** Making the AI assistant genuinely useful on a real fault code, not just a generic chatbot bolted onto the site.
- **Solution:** Wrote a structured system prompt listing real fault codes and scenario-based safety instructions, so responses are grounded in the actual hazards technicians face rather than generic troubleshooting advice.
- **Challenge:** My dad isn't technical, so the admin experience needed to be something he could use without help.
- **Solution:** Built every admin action (editing prices, managing users, logging maintenance) as a simple form inside the same dashboard, instead of a separate technical panel.

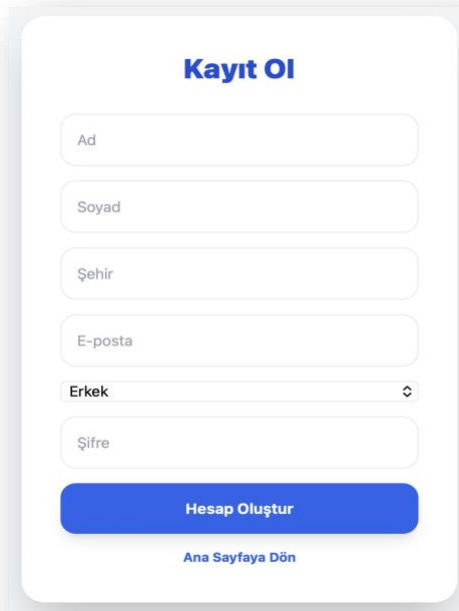
Project Visuals & Output Showcase

User Onboarding



The login screen features a white card with rounded corners on a light gray background. At the top, the title "Giriş Yap" is in bold blue, with the subtitle "YÖNETİM PANELİNE ERİŞİN" below it. A close button (X) is in the top right. The form includes an email input field with "erimakcan@gmail.com", a password input field with seven dots, and a blue "Sisteme Gir" button. Below the button, a link "HENÜZ HESABINIZ YOK MU? ŞİMDİ KAYIT OL" is displayed. At the bottom, a link "← ANASAYFA" is shown.

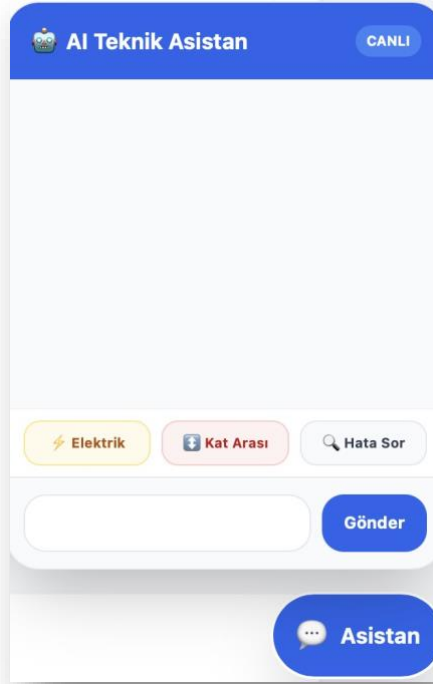
Login screen for returning users.



The registration screen features a white card with rounded corners on a light gray background. The title "Kayıt Ol" is in bold blue. The form includes input fields for "Ad", "Soyad", "Şehir", and "E-posta". A dropdown menu for "Erkek" is shown with a downward arrow. A password input field with "Şifre" is also present. A blue "Hesap Oluştur" button is at the bottom. Below the button, a link "Ana Sayfaya Dön" is shown.

Registration screen for new accounts, created with the default "user" role.

AI Technical Assistant



The assistant at rest, with quick-access buttons for the most common scenarios: power outage, stuck between floors, or looking up a fault code.

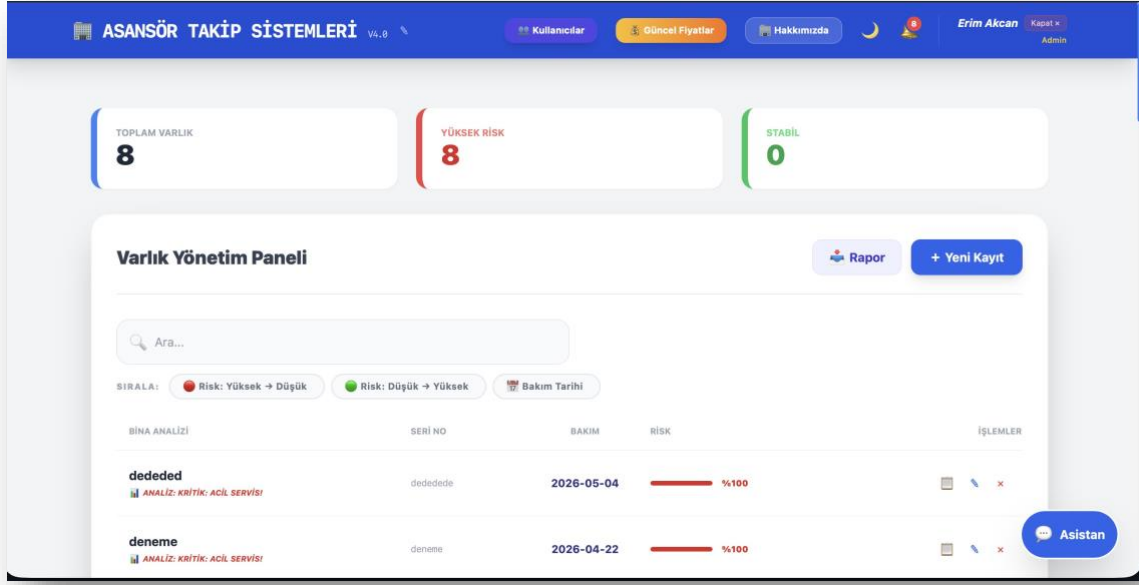


Explaining fault code E05, including which safety step to take before calling for help.

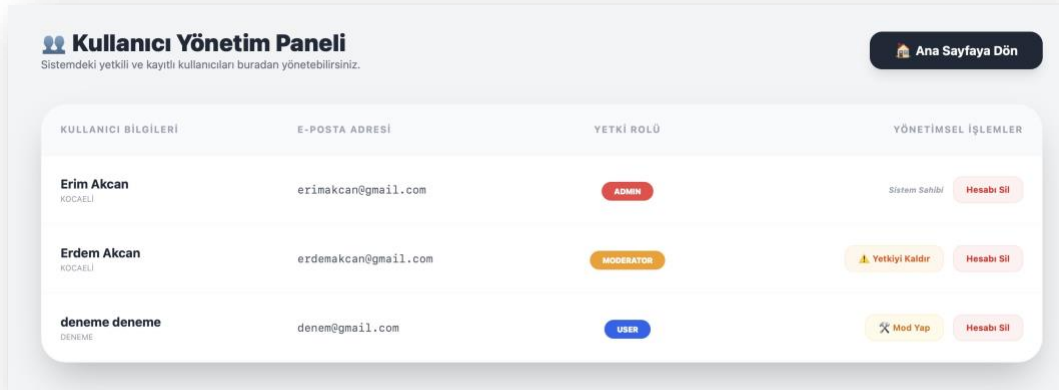


Walking a technician through a power outage scenario, telling them to wait for the automatic rescue system rather than trying to open the doors manually.

Admin Dashboard



Every tracked elevator, sorted by risk, with a live count of assets and high-risk cases.



The screenshot shows the 'Kullanıcı Yönetim Paneli' (User Management Panel). It includes a header with the title and a description: 'Sistemdeki yetkili ve kayıtlı kullanıcıları buradan yönetebilirsiniz.' (You can manage the authorized and registered users in the system from here). There is a button 'Ana Sayfaya Dön' (Return to Home Page). The main content is a table with columns: 'KULLANICI BİLGİLERİ', 'E-POSTA ADRESİ', 'YETKİ ROLÜ', and 'YÖNETİMSSEL İŞLEMLER'. The table lists three users: 'Erim Akcan' (KOCALI) with role 'ADMIN', 'Erdem Akcan' (KOCALI) with role 'MODERATOR', and 'deneme deneme' (DENEME) with role 'USER'. Each user row has buttons for 'Yetkiyi Kaldır' (Remove Permission) and 'Hesabı Sil' (Delete Account). The 'deneme deneme' row also has a 'Mod Yap' (Make Mod) button.

KULLANICI BİLGİLERİ	E-POSTA ADRESİ	YETKİ ROLÜ	YÖNETİMSSEL İŞLEMLER
Erim Akcan KOCALI	erimakcan@gmail.com	ADMIN	Sistem Sahibi Hesabı Sil
Erdem Akcan KOCALI	erdemakcan@gmail.com	MODERATOR	Yetkiyi Kaldır Hesabı Sil
deneme deneme DENEME	denem@gmail.com	USER	Mod Yap Hesabı Sil

Role management panel: admin, moderator, and user accounts, with role changes handled from the UI.

GÜNCEL FİYATLARIMIZ

Hizmet Fiyat Listesi

Kapı Değişimi1000

ADMIN PANELİ — DÜZENLEME

Kapı Değişimi1000Kaydet

Yeni başlık (Orn: Kapı Değişimi)Fiyat (Orn: 1500)+ Ekle

Fiyatlar bilgilendirme amaçlıdır. Kesin teklif için bizimle iletişime geçin.

Admin editing the public service price list directly, no code changes required.

dededed — Bakım Kaydı

Seri No: dededede

YAPILAN İŞLEMLER

☐ Genel Bakım

☐ Motor Yağ Değişimi

☐ Ampül Değişimi

☐ Kapı Düzeltmesi

☐ Kablo Kontrolü

☐ Fren Ayarı

☐ Ray Yağlama

☐ Güvenlik Testi

GEÇMİŞ BAKIM KAYITLARI

21 MAYIS 2026

14:55 — Erim Akcan

Genel Bakım

Tamamlandı

21 MAYIS 2026

14:55 — Erim Akcan

Motor Yağ Değişimi

Tamamlandı

19 MAYIS 2026

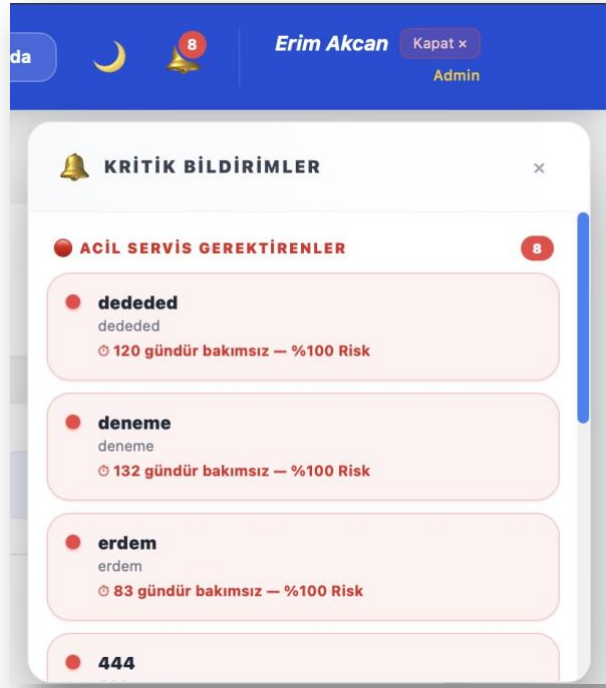
12:04 — Erdem Akcan

Genel Bakım

Motor Yağ Değişimi

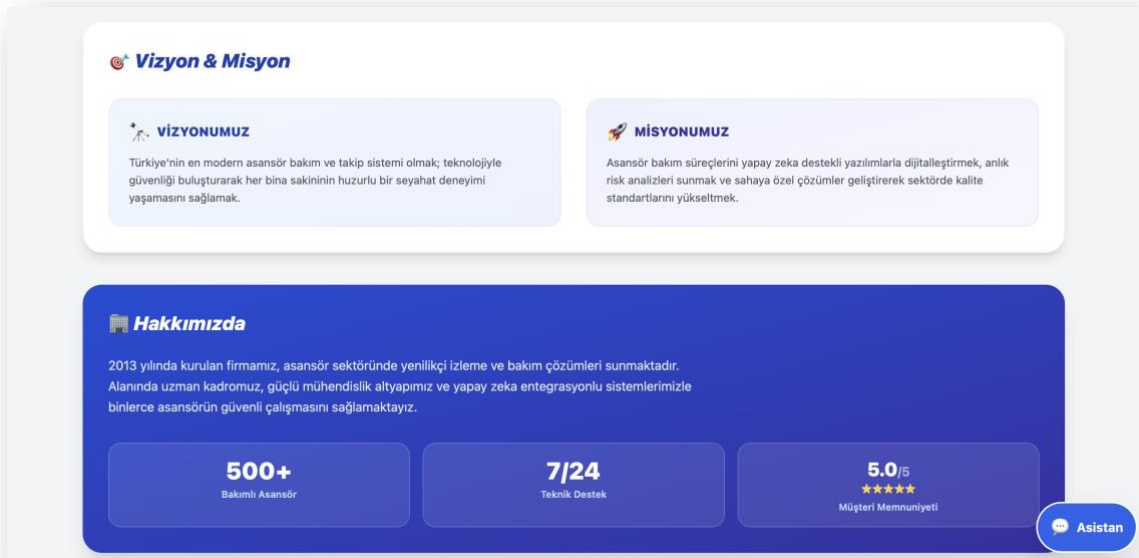
Tamamlandı

Maintenance record for a single elevator, showing completed tasks and full service history.



Critical alerts surfacing every elevator overdue for maintenance, with days-since-service and risk level.

Public-Facing Content



The vision, mission, and about section on the public site, also editable from the admin panel as key-value content, not hardcoded into the page.

Current Status & Roadmap

The platform is in active use for tracking my dad's elevators, logging maintenance, and answering fault code questions through the AI assistant. It's currently private since it's a live tool for his business and still evolving. Once it's further along, I'm planning to open the repository up publicly on GitHub.